CryptoEx

Backend Concepts and Interview Guide

A detailed explanation of the architecture, concepts, tradeoffs, and interview talking points behind the CryptoEx centralized exchange project.

Read this when you want to explain the project confidently in interviews: what each concept means, why it exists, and exactly how CryptoEx uses it.

Project type	Production-style centralized crypto exchange
Core systems	Auth, KYC, wallets, ledger, custody, trading, market data, payments
Backend stack	Node.js, Express, PostgreSQL, Prisma, Redis, BullMQ, Kafka
Integrations	BitGo test custody, Binance WebSocket streams, Razorpay test checkout
Deployment	AWS EC2, Docker, NGINX, PM2, GitHub Actions

How to Use This Guide

This guide is intentionally detailed. It is written for interview preparation, not just project documentation. The pattern is simple: first understand the concept in plain language, then understand why a backend system needs it, then learn how CryptoEx implemented it, then practice the interview answer.

Do not describe CryptoEx as a real-money exchange. The accurate phrase is production-style exchange using testnet and sandbox integrations. That honesty makes the project sound more mature, not less impressive.

Fast project pitch

CryptoEx is a centralized crypto exchange backend. It includes secure authentication, TOTP 2FA, KYC/admin workflows, BitGo custody deposits and withdrawals, an internal wallet ledger, spot order matching, Binance market data projections, Razorpay test deposits, Redis rate limiting, BullMQ workers, Kafka event streams, CI/CD, and AWS deployment.

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1. System Overview

CryptoEx is modeled like a centralized exchange. Users deposit crypto into exchange-controlled custody addresses. After the deposit is confirmed, CryptoEx credits an internal balance. Trading happens inside the exchange database and ledger. Withdrawals move internal balances back out to a blockchain address through the custody provider.

```
User deposit -> BitGo custody address -> webhook -> worker -> deposit record -> ledger credit
User order -> validation -> funds locked -> matching engine -> trade rows -> ledger settlement
User withdrawal -> security check -> funds locked -> worker submits transfer -> final status
```

This separation is the key design idea. Blockchain is used for entering and leaving the exchange. The internal ledger is used for fast exchange activity.

Controller-service-worker separation

What it means

A controller handles HTTP input and output. A service owns business rules. A worker handles async jobs outside the request-response path.

Why it is used

This makes the code easier to test, reason about, and reuse. A worker can call the same service logic as an API route without duplicating financial rules.

How CryptoEx uses it

CryptoEx has routers and controllers for HTTP endpoints, services for wallet/trading/custody/payment logic, and workers for webhooks, withdrawals, market data, candles, and outbox publishing.

Interview answer

> *I separated HTTP handling from business logic. Controllers stay thin, services own rules, and workers handle slow or retryable tasks.*

2. Authentication and Sessions

JWT access tokens

What it means

A JWT access token is a signed token proving that a user authenticated. It usually expires quickly.

Why it is used

Short expiry limits damage if it leaks and keeps protected API calls fast because the backend can verify the signature.

How CryptoEx uses it

CryptoEx uses access tokens for authenticated routes such as wallet balances, orders, KYC, profile, and payment APIs.

Interview answer

> *Access tokens are short-lived proof of login. They authorize normal protected API calls.*

Refresh tokens

What it means

A refresh token is a longer-lived token used to obtain a new access token.

Why it is used

It lets users stay logged in without making access tokens long-lived. It also creates a place for rotation and revocation.

How CryptoEx uses it

CryptoEx uses refresh-token based sessions and Redis-backed session state.

Interview answer

> *Access tokens are for short-lived API access. Refresh tokens are for session continuity and can be rotated or revoked.*

HTTP-only cookies

What it means

An HTTP-only cookie cannot be read by frontend JavaScript.

Why it is used

It reduces token theft risk from XSS compared with localStorage.

How CryptoEx uses it

CryptoEx uses cookies for browser authentication between the frontend and API.

Interview answer

> *HTTP-only cookies let the browser send tokens automatically without exposing them to JavaScript.*

Redis-backed sessions

What it means

Redis stores temporary server-side session metadata.

Why it is used

JWTs are stateless, but real apps need logout, revoke, expiry, and active-session control.

How CryptoEx uses it

CryptoEx stores session information in Redis so refresh flows and logout can be controlled server-side.

Interview answer

> *JWT gives fast verification. Redis gives session control. Together they are practical for secure web sessions.*

3. 2FA, Step-Up, and Trusted Devices

TOTP 2FA

What it means

TOTP means Time-based One-Time Password. An authenticator app and backend share a secret and independently generate the same short code for the current time window.

Why it is used

Passwords can be stolen. TOTP adds a second factor that an attacker usually does not have.

How CryptoEx uses it

CryptoEx supports setup QR codes, encrypted 2FA secrets, login challenges, and recent verification.

Interview answer

> *TOTP proves the user has the authenticator secret, not just the password.*

Encrypted 2FA secret

What it means

The TOTP secret must be stored so the backend can verify future codes, but it should not be stored in plain text.

Why it is used

If the database leaks, plain 2FA secrets would let attackers generate OTP codes.

How CryptoEx uses it

CryptoEx encrypts 2FA secrets using AES-256-GCM.

Interview answer

> *The backend needs the secret to verify OTP, so encryption at rest is the correct protection.*

Recent 2FA step-up

What it means

Step-up means asking for fresh OTP verification before sensitive actions.

Why it is used

A logged-in session may be old or compromised. Withdrawals and security changes need fresh proof.

How CryptoEx uses it

CryptoEx has recent 2FA checks for sensitive wallet/security operations.

Interview answer

> *Login proves who started the session. Step-up proves the user is present right now.*

Trusted devices

What it means

A trusted device is a remembered device that can reduce repeated login OTP prompts.

Why it is used

It improves UX while still letting users revoke unfamiliar devices.

How CryptoEx uses it

CryptoEx stores trusted-device records and exposes list/delete APIs.

Interview answer

> *Trusted devices are convenience, not a replacement for 2FA. Users can review and revoke them.*

4. KYC and Admin Workflows

KYC

What it means

KYC means Know Your Customer: collecting and reviewing identity information.

Why it is used

Financial apps need KYC for compliance, fraud prevention, and risk controls.

How CryptoEx uses it

CryptoEx lets users submit identity details and documents, then admins review, approve, or reject submissions.

Interview answer

> *I modeled KYC as a lifecycle with submission, document upload, admin review, rejection reasons, and audit history.*

RBAC

What it means

RBAC means Role-Based Access Control. A user role controls what actions are allowed.

Why it is used

Admin actions must not be reachable by normal users. Authentication alone is not enough.

How CryptoEx uses it

CryptoEx uses USER and ADMIN roles and protects admin routes with authentication plus admin middleware.

Interview answer

> *Authentication answers who the user is. Authorization answers what they can do.*

5. Wallets and Balances

Internal exchange wallet

What it means

An internal wallet is a database wallet representing user balances inside the exchange.

Why it is used

Centralized exchanges cannot settle every trade on-chain. Trading must be fast and cheap, so exchanges use internal balances.

How CryptoEx uses it

CryptoEx credits internal balances after deposits and updates them through ledger entries during trading and withdrawals.

Interview answer

> *Blockchain handles deposits and withdrawals. The exchange ledger handles fast internal trading.*

Available balance

What it means

Available balance is spendable balance.

Why it is used

Users should only spend or withdraw money that is not already reserved.

How CryptoEx uses it

CryptoEx uses AVAILABLE wallet accounts for funds that can be used now.

Interview answer

> *Available means free to spend.*

Locked balance

What it means

Locked balance is reserved for a pending operation.

Why it is used

It prevents double spending. The same USDT cannot be used for an open buy order and a withdrawal at the same time.

How CryptoEx uses it

CryptoEx locks quote assets for buy orders, base assets for sell orders, and assets for withdrawals.

Interview answer

> *Locked means reserved until the order fills/cancels or the withdrawal completes/fails.*

6. Double-Entry Ledger

Double-entry accounting

What it means

Every financial movement is recorded with matching debit and credit entries. Total debits equal total credits.

Why it is used

It creates auditability and financial correctness. You can explain where money came from and where it went.

How CryptoEx uses it

CryptoEx uses ledger transactions and entries for deposits, withdrawals, locks, unlocks, trade settlement, and adjustments.

Interview answer

> *Double-entry is the financial safety layer. Instead of just changing balance numbers, every movement has an auditable trail.*

Ledger transaction vs ledger entry

What it means

A ledger transaction is the business event. Ledger entries are the debit/credit lines inside it.

Why it is used

One business event can move value across multiple accounts. Grouping entries makes the event understandable.

How CryptoEx uses it

CryptoEx groups entries under transaction types such as DEPOSIT, WITHDRAWAL, FUNDS_LOCK, FUNDS_UNLOCK, and TRADE_SETTLEMENT.

Interview answer

> *The transaction says what happened. The entries show exactly which accounts moved.*

Why keep balance rows

What it means

Balance rows are fast current snapshots.

Why it is used

Calculating every balance by summing all ledger entries on each request would be slow.

How CryptoEx uses it

CryptoEx keeps wallet balances for reads but updates them through ledger operations.

Interview answer

> *The balance is the snapshot. The ledger is the history explaining the snapshot.*

7. BitGo Custody

Custody provider

What it means

A custody provider manages blockchain wallets, signing, transfer broadcasting, and wallet security.

Why it is used

Building secure custody from scratch is risky. Exchanges usually integrate custody infrastructure.

How CryptoEx uses it

CryptoEx uses BitGo test custody for receive addresses, transfer fetching, webhooks, and withdrawals.

Interview answer

> *BitGo handles wallet infrastructure. CryptoEx handles product logic, user balances, and accounting.*

Receive address

What it means

A receive address is where a user sends crypto.

Why it is used

Unique addresses help map incoming transfers to the correct user and network.

How CryptoEx uses it

CryptoEx creates or fetches BitGo receive addresses and stores them against user, asset, and network.

Interview answer

> *The deposit address tells the backend which user should be credited.*

Testnet

What it means

A testnet is a blockchain network for testing. Tokens have no real value.

Why it is used

It lets developers test blockchain flows without risking real funds.

How CryptoEx uses it

CryptoEx uses BTC testnet and ETH Hoodi style test flows.

Interview answer

> *Testnets prove the architecture safely before any real-money system.*

8. Deposits

Deposits are complex because the backend depends on external blockchain and custody state. A correct deposit flow must be idempotent, confirmation-aware, and recoverable.

```
Deposit address requested
-> user sends crypto
-> BitGo webhook arrives
-> API stores webhook event
-> BullMQ worker fetches official transfer
-> deposit record is created/updated
-> confirmations checked
-> ledger credits user balance
-> deposit status becomes CREDITED
```

Confirmations

What it means

Confirmations count blocks added after the transaction.

Why it is used

More confirmations reduce risk from chain reorgs and unreliable finality.

How CryptoEx uses it

CryptoEx stores required confirmations per asset/network and credits when the requirement is met.

Interview answer

> *A webhook is not enough. The backend should verify provider state and confirmation count before crediting.*

Deposit finalizer recovery

What it means

Recovery scans deposits that were detected but not fully credited.

Why it is used

Workers can restart and providers can fail temporarily. Recovery prevents funds from staying stuck.

How CryptoEx uses it

CryptoEx added startup recovery for pending BitGo deposit finalizations.

Interview answer

> *Recovery is a safety net for asynchronous financial workflows.*

9. Withdrawals

Withdrawals are high risk because funds leave the platform. They need authentication, security checks, balance locking, provider submission, status tracking, failure handling, and reconciliation.

```
Withdrawal request
-> validate asset/network/address/amount
-> auth and security checks
-> lock available funds
-> queue worker
-> submit BitGo transfer
-> track provider status
-> complete or unlock funds
```

Async withdrawals

What it means

Withdrawals involve external providers and blockchain state, so they can take time.

Why it is used

The HTTP request should not wait for provider finality. Workers can retry and update status later.

How CryptoEx uses it

CryptoEx uses BullMQ workers to submit and finalize withdrawals.

Interview answer

> *The API starts the withdrawal. Workers finish it safely in the background.*

Unlock on failure

What it means

If a withdrawal fails, locked funds must return to available.

Why it is used

Otherwise the user loses access even though no successful transfer happened.

How CryptoEx uses it

CryptoEx models fund locking and unlocking through ledger entries.

Interview answer

> *Locking prevents double spend. Unlocking protects the user on failure.*

10. Razorpay Fiat Test Deposits

Fiat

What it means

Fiat means government-issued money like INR or USD.

Why it is used

Many exchanges allow users to add money through payment gateways or bank rails.

How CryptoEx uses it

CryptoEx uses Razorpay test mode to simulate INR deposits and credit test USDT.

Interview answer

> *Fiat is normal money. In this project, Razorpay test checkout simulates adding INR and credits test USDT.*

Razorpay order

What it means

A Razorpay order is a server-created payment intent.

Why it is used

The backend must control amount and currency so the browser cannot fake them.

How CryptoEx uses it

CryptoEx creates a FiatDeposit row and Razorpay order before checkout.

Interview answer

> *The payment order starts on the backend, not the frontend.*

Signature verification

What it means

The backend recomputes the expected Razorpay signature and compares it with the returned signature.

Why it is used

It proves the payment response was not forged by the browser.

How CryptoEx uses it

CryptoEx verifies `orderId|paymentId` using HMAC SHA-256 and timing-safe comparison.

Interview answer

> *Never credit funds just because frontend says payment succeeded. Verify provider signature first.*

11. Trading Pairs and Orders

Trading pair

What it means

A trading pair defines the base and quote assets, like BTC-USDT.

Why it is used

It controls what asset is bought/sold, decimal precision, minimum size, and active status.

How CryptoEx uses it

CryptoEx stores trading pairs with base asset, quote asset, price decimals, quantity decimals, and minimum quote amount.

Interview answer

> *In BTC-USDT, BTC is base and USDT is quote. Buy means spend USDT to receive BTC.*

Limit order

What it means

A limit order says buy or sell at a specific price or better.

Why it is used

It gives users price control and can rest in the order book.

How CryptoEx uses it

CryptoEx supports limit orders and locks funds before matching.

Interview answer

> *A buy limit says do not pay more than this price. A sell limit says do not sell below this price.*

Partial fill

What it means

A partial fill means only part of the order matched.

Why it is used

Large orders often match against several smaller orders.

How CryptoEx uses it

CryptoEx tracks filled quantity, remaining quantity, and PARTIALLY_FILLED status.

Interview answer

> *Partial fills are normal in order books; the remaining quantity can stay open.*

12. Matching Engine

Price-time priority

What it means

Best price wins first, and for the same price the oldest order wins first.

Why it is used

It is a fair and common exchange matching rule.

How CryptoEx uses it

CryptoEx sorts eligible opposite orders by price and creation time.

Interview answer

> *For buys, match the lowest eligible sells first. For sells, match the highest eligible buys first. Ties use order age.*

Maker and taker

What it means

Maker is the resting order. Taker is the incoming order that matches it.

Why it is used

Fees and analytics often distinguish liquidity maker and consumer.

How CryptoEx uses it

CryptoEx records maker/taker order IDs and sides on trade rows.

Interview answer

> *Maker made liquidity. Taker consumed it.*

Execution price

What it means

Execution price is the price used for the trade.

Why it is used

Most exchanges execute at the resting maker price.

How CryptoEx uses it

CryptoEx uses the opposite/resting order price as the trade price.

Interview answer

> *If a buy at 2600 matches a resting sell at 2500, the trade can execute at 2500.*

Atomic settlement

What it means

Atomic settlement means trade record, balance movement, and order updates commit together.

Why it is used

Financial systems cannot allow half-finished trades.

How CryptoEx uses it

CryptoEx performs settlement in database transactions.

Interview answer

> *A trade is only valid if trade row, ledger entries, and order statuses commit together.*

13. Order Book, Trades, and Candles

Order book

What it means

An order book lists open buy and sell interest grouped by price.

Why it is used

Users need it to understand depth, spread, and liquidity.

How CryptoEx uses it

CryptoEx exposes bids, asks, quantity, totals, and order counts.

Interview answer

> *Bids are buyers. Asks are sellers. Spread is best ask minus best bid.*

Recent trades

What it means

Recent trades show completed executions.

Why it is used

They show what actually happened, while order book shows pending intent.

How CryptoEx uses it

CryptoEx exposes recent trades with price, quantity, quote amount, maker/taker side, and timestamps.

Interview answer

> *Order book shows intent. Recent trades show execution.*

OHLCV candles

What it means

Candles summarize open, high, low, close, and volume for a time interval.

Why it is used

Charts need compressed time-series data instead of every raw trade forever.

How CryptoEx uses it

CryptoEx stores exchange-trade candles and reference-market candles.

Interview answer

> *A 1-minute candle summarizes the market during that minute.*

14. Market Data

Reference market data

What it means

Reference data comes from an external market such as Binance.

Why it is used

A new exchange may have few internal trades, so charts would be empty without reference market movement.

How CryptoEx uses it

CryptoEx streams Binance tickers and klines, publishes events to Kafka, and projects candles/tickers for APIs.

Interview answer

> *Reference market data lets the app show realistic price movement while internal liquidity grows.*

Market projection

What it means

A projection is a read-optimized view built from events.

Why it is used

Consumers need fast reads without recalculating from raw events every time.

How CryptoEx uses it

CryptoEx projects market events into Redis and PostgreSQL candle tables.

Interview answer

> *Events are the stream; projections are fast query views.*

Backfill

What it means

Backfill loads historical data.

Why it is used

Stream workers only collect data while running. Backfill fills older chart ranges.

How CryptoEx uses it

CryptoEx has a market candle backfill script.

Interview answer

> *Streaming handles future data. Backfill handles the past.*

15. Kafka and Outbox

Kafka

What it means

Kafka is a durable event log where producers write events and consumers read them independently.

Why it is used

It decouples systems and lets multiple consumers react to the same event stream.

How CryptoEx uses it

CryptoEx uses Kafka for market, trade, and domain events.

Interview answer

> *Kafka is useful when many parts of the system need the same event without tight coupling.*

Topic

What it means

A Kafka topic is a named event stream.

Why it is used

Topics organize events by category.

How CryptoEx uses it

CryptoEx uses topics for market and trade/domain events.

Interview answer

> *A topic is like a durable channel for a type of event.*

Transactional outbox

What it means

The outbox stores an event in the database in the same transaction as the business change.

Why it is used

It solves the dual-write problem where DB success and Kafka publish success can get out of sync.

How CryptoEx uses it

CryptoEx writes outbox rows, then an outbox worker publishes them to Kafka and marks them published.

Interview answer

> *Outbox makes event publishing reliable because the event is committed with the business data first.*

16. BullMQ and Workers

BullMQ

What it means

BullMQ is a Redis-backed job queue for Node.js.

Why it is used

It is good for retryable background jobs where one worker should process a task.

How CryptoEx uses it

CryptoEx uses BullMQ for custody webhook jobs and withdrawal jobs.

Interview answer

> *Kafka is for event streams. BullMQ is for background jobs with retries.*

Worker process

What it means

A worker is a separate process that handles async work.

Why it is used

Workers keep slow tasks away from the HTTP request path and can restart independently.

How CryptoEx uses it

CryptoEx runs custody, withdrawal, outbox, trade-candle, market-data, and market-projection workers.

Interview answer

> *The API accepts work. Workers finish slow or retryable work.*

17. Redis

Redis

What it means

Redis is a fast in-memory data store.

Why it is used

It is great for temporary state, counters, cache, queues, and sessions.

How CryptoEx uses it

CryptoEx uses Redis for sessions, BullMQ, rate limiting, and market cache.

Interview answer

> *Redis handles fast temporary state. PostgreSQL remains the durable source of truth.*

Why not store balances in Redis only

What it means

Redis is not the ideal durable financial database for this architecture.

Why it is used

Balances need durable transactions, relations, indexes, and audit history.

How CryptoEx uses it

CryptoEx stores financial state in PostgreSQL and uses Redis as supporting infrastructure.

Interview answer

> *Use Redis for speed, not as the only accounting record.*

18. Rate Limiting

Sliding-window counter

What it means

A sliding-window counter estimates request count across current and previous windows.

Why it is used

It avoids fixed-window boundary bursts while staying cheaper than storing every request timestamp.

How CryptoEx uses it

CryptoEx uses it for login, signup, password reset, 2FA, KYC, withdrawal, deposit address, and admin actions.

Interview answer

> *Sliding window is a good security limiter for sensitive actions.*

Token bucket

What it means

A token bucket has capacity and refills over time. Requests spend tokens.

Why it is used

It allows bursts but limits sustained traffic.

How CryptoEx uses it

CryptoEx uses token bucket limiting for market and trading read APIs.

Interview answer

> *Token bucket is good for read-heavy APIs because users can burst briefly but not overload the system forever.*

Redis Lua scripts

What it means

Lua scripts run atomically inside Redis.

Why it is used

Rate limiters must read, calculate, update, and expire counters safely under concurrency.

How CryptoEx uses it

CryptoEx uses Redis Lua scripts for both limiter types.

Interview answer

> *Lua prevents race conditions in the rate-limit decision.*

19. Webhook Security and Idempotency

Webhook

What it means

A webhook is an HTTP callback from an external provider.

Why it is used

Providers use it to notify your backend about transfers or payment updates.

How CryptoEx uses it

BitGo calls CryptoEx at `/webhooks/custody/bitgo`.

Interview answer

> *Webhooks are external events, so they must be authenticated and validated.*

Raw body

What it means

Some signatures are calculated over the exact raw request body.

Why it is used

Parsing JSON can change formatting and break signature verification.

How CryptoEx uses it

CryptoEx captures raw body through Express JSON verify.

Interview answer

> *For signed webhooks, raw body handling is important because the signature depends on exact bytes.*

Replay protection

What it means

Replay protection rejects repeated or old webhook attempts.

Why it is used

An attacker might resend a valid signed request.

How CryptoEx uses it

CryptoEx uses replay protection and webhook event storage.

Interview answer

> *A valid signature is not enough; the system must also reject duplicate events.*

Idempotency

What it means

Idempotency means repeated attempts produce the same final result without duplicate side effects.

Why it is used

External providers and clients retry. Financial systems must not double credit or double withdraw.

How CryptoEx uses it

CryptoEx uses provider IDs, unique keys, and status checks around deposits, trades, and events.

Interview answer

> *Idempotency is mandatory in financial systems.*

20. Database Design

Prisma schema

What it means

The Prisma schema defines models, enums, relations, and indexes.

Why it is used

It makes the database structure explicit and generates the Prisma client.

How CryptoEx uses it

CryptoEx models users, KYC, wallets, balances, ledger, deposits, withdrawals, trading pairs, orders, trades, candles, fiat deposits, and outbox events.

Interview answer

> *The schema is the real map of the backend business state.*

Indexes

What it means

Indexes speed up common lookups.

Why it is used

Trading and wallet pages query by user, asset, status, pair, and time.

How CryptoEx uses it

CryptoEx uses indexes for user IDs, order IDs, trading pair IDs, and timestamps.

Interview answer

> *Indexes make reads faster but add write cost, so they should match actual query patterns.*

Transactions

What it means

A transaction groups DB operations so they all commit or roll back together.

Why it is used

Financial updates cannot be half-complete.

How CryptoEx uses it

CryptoEx uses transactions for ledger movements, settlements, deposits, and withdrawals.

Interview answer

> *Transactions protect correctness when multiple rows must change together.*

Row locking hardening

What it means

Row locks prevent concurrent transactions from incorrectly changing the same hot rows.

Why it is used

High-volume matching and wallet updates need explicit concurrency control.

How CryptoEx uses it

CryptoEx has transactional flows, and explicit row locking or serializable isolation is a planned hardening step.

Interview answer

> *I would say: the project uses transactions, and the next production hardening step is explicit locks or serializable isolation on hot balance and order rows.*

21. Deployment and CI/CD

NGINX

What it means

NGINX is a reverse proxy in front of the Node API.

Why it is used

It handles public HTTP/HTTPS traffic and forwards requests to the internal app port.

How CryptoEx uses it

CryptoEx uses NGINX for the API domain.

Interview answer

> *Node runs internally, NGINX exposes it publicly.*

PM2

What it means

PM2 is a Node process manager.

Why it is used

It restarts processes, keeps workers alive, and saves process lists across reboots.

How CryptoEx uses it

CryptoEx runs API and worker scripts with PM2.

Interview answer

> *PM2 is practical for supervising Node processes on a single EC2 instance.*

Docker

What it means

Docker runs infrastructure consistently.

Why it is used

PostgreSQL, Redis, Kafka, and Kafka UI can run the same way locally and on EC2.

How CryptoEx uses it

CryptoEx uses Docker compose for infrastructure services.

Interview answer

> *Docker makes local and server infrastructure easier to reproduce.*

GitHub Actions CI/CD

What it means

CI/CD automatically validates and deploys code.

Why it is used

It reduces manual deployment mistakes and catches broken code before production.

How CryptoEx uses it

CryptoEx backend workflow validates Prisma and JS syntax, deploys to EC2, runs migrations, restarts PM2 processes, and runs smoke tests.

Interview answer

> *CI checks quality. CD deploys repeatably and verifies the app is alive after deployment.*

22. Benchmarks

Benchmarks should be described carefully. These are local development measurements, not guaranteed production capacity. They are still useful because they show you measured the system instead of guessing.

- Trading order creation: 30 limit orders at concurrency 3; warm runs around 150 orders/sec with p99 around 40ms.

- Wallet balances: 5 requests/sec for 20 seconds; avg 17.24ms, p50 11ms, p99 122ms.

- Order book: 5 requests/sec for 20 seconds; avg 7.88ms, p50 5ms, p99 47ms.

- Recent trades: 5 requests/sec for 20 seconds; avg 9.77ms, p50 5ms, p99 57ms.

- Market overview: 5 requests/sec for 20 seconds; avg roughly 203-490ms depending on cache/provider state.

- Rate limiter stress: around 16.9k requests/sec handled while rejecting excess traffic as intended.

- Blockchain health: avg around 541ms, p99 around 1.96s because it checks external/RPC health.

> *Interview-safe wording: I ran local endpoint benchmarks to validate obvious bottlenecks and rate-limit behavior. I would not present them as production capacity. For production numbers, I would run load tests in a staging environment with production-like database, Redis, Kafka, workers, and observability.*

23. Production Hardening Roadmap

- Client-facing WebSocket gateway for live order book, trades, tickers, balances, and order updates.

- Explicit row locking or serializable isolation for high-concurrency wallet and matching rows.

- Automated unit and integration tests for ledger invariants, matching edge cases, webhooks, payments, and withdrawals.

- Centralized observability with metrics, alerts, traces, worker lag, and dead-letter dashboards.

- Staging environment mirroring production before deployment.

- Admin reconciliation dashboard comparing custody provider state, internal ledger state, pending deposits, and stuck withdrawals.

- Secret rotation, dependency scanning, security review, and incident response runbooks.

This roadmap is important. It shows maturity because you can explain what is done and what still needs production

24. Interview Questions and Answers

Q: Why did you use an internal ledger instead of direct blockchain balances?

> *Because centralized exchanges need fast internal trading. Blockchain transfers are slow and expensive for every trade. Deposits and withdrawals touch the chain; trades update internal balances and ledger entries.*

Q: What happens when a user places a buy order?

The backend validates pair, price, quantity, and minimums. It calculates quote required, locks quote funds, creates the order, matches eligible sell orders, records trades, updates order status, and settles through ledger entries.

Q: How do you prevent double crediting a deposit?

Use idempotency. Store provider transfer IDs and webhook events uniquely, check existing records before crediting, and make ledger crediting happen once.

Q: Why BullMQ and Kafka both?

BullMQ is for retryable jobs where one worker processes a task. Kafka is for durable event streams where many consumers can react independently.

Q: What is the outbox pattern?

It stores an event in the database in the same transaction as the business update. A worker publishes it to Kafka later. This avoids DB success but event publish failure.

Q: What is price-time priority?

Best price matches first. If prices are equal, the older order matches first. That is fair and common for exchanges.

Q: What is maker and taker?

Maker is the resting order already in the book. Taker is the incoming order that matches against it.

Q: How do you verify Razorpay payments?

The backend recomputes the HMAC signature from Razorpay order ID and payment ID using the secret, then compares it safely before crediting funds.

Q: What are the biggest next improvements?

WebSocket gateway, explicit concurrency locking, automated tests, observability, staging, and admin reconciliation.

Q: How would you scale the project?

Run multiple API and worker instances, move DB/Redis/Kafka to managed services, add read replicas for public reads, add WebSocket gateways, instrument worker lag, and use staging load tests.

Final Project Pitch

CryptoEx is a production-style centralized exchange backend that connects real backend systems together: authentication, 2FA, KYC, custody, internal accounting, trading, market data, payments, workers, events, deployment, and CI/CD. The impressive part is not one endpoint. It is the complete system shape and the way financial correctness, async processing, provider integrations, and user-facing APIs fit together.

Short pitch: I built CryptoEx to understand how real exchanges work beyond CRUD APIs. It includes secure auth, TOTP 2FA, KYC, BitGo custody, deposits and withdrawals, double-entry wallet accounting, a spot matching engine, Binance-powered market data, Razorpay test deposits, Redis rate limiting, BullMQ workers, Kafka event streams, CI/CD, and AWS deployment. I can explain the tradeoffs behind each part and what I would harden next for real production usage.